

THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF PHYSICS

FINAL EXAMINATION

PHYS3114 – Electrodynamics

Term 3, 2023

1. Answer ALL questions
2. ALL QUESTIONS ARE OF EQUAL VALUE.
Marks available for each question are shown in the examination paper.
3. Answer each question in a separate book.
4. University-approved calculators may be used.
5. All answers must be written in ink. Except where they are expressly required, pencils may only be used for drawing, sketching or graphical work.
6. This paper may be retained by the candidate.

PHYS3114 — Definitions and Formulae

$$F(s) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x s} dx \quad (1)$$

$$\mathcal{F}\{f(x)\} = 2 \int_0^\infty f_e(x) \cos(2\pi s x) dx - 2i \int_0^\infty f_o(x) \sin(2\pi s x) dx \quad (2)$$

$$\gamma(P_1, P_2) = \frac{\iint_{\text{source}} I(\zeta, \eta) e^{\frac{-2\pi i}{\lambda}(p\zeta + q\eta)} d\zeta d\eta}{\iint_{\text{source}} I(\zeta, \eta) d\zeta d\eta} \quad (3)$$

$$V(x', y') = C_1 C_2 \lambda^2 f_o^2 \iint_{\text{bfp}} F\left(\frac{\zeta}{\lambda f_o}, \frac{\eta}{\lambda f_o}\right) e^{-i2\pi(\frac{f_o}{D}x' \frac{\zeta}{\lambda f_o} + \frac{f_o}{D}y' \frac{\eta}{\lambda f_o})} d\left(\frac{\zeta}{\lambda f_o}\right) d\left(\frac{\eta}{\lambda f_o}\right) \quad (4)$$

$$\mathcal{F}\left\{\frac{1}{2}\delta(r-a)\right\} = \pi a J_0(2\pi a \rho) \quad (5)$$

$$\mathcal{F}\left\{\prod\left(\frac{r}{2a}\right)\right\} = a \frac{J_1(2\pi a \rho)}{\rho} \quad (6)$$

$$F(\rho) = 2\pi \int_0^\infty f(r) J_0(2\pi \rho r) r dr \quad (7)$$

$$\frac{m\lambda}{2} = \frac{r_m^2}{2} \left(\frac{1}{r_0} + \frac{1}{b_0} \right) \quad (8)$$

$$\frac{d^2 P}{dt^2} + \gamma \frac{dP}{dt} + \omega_0^2 P = \frac{e^2 N}{m} \left(E + \frac{\nu P}{\epsilon_0} \right). \quad (9)$$

$$\hat{K} - 1 = \frac{\omega_p^2(\omega_{0'}^2 - \omega^2)}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} + i \frac{\omega_p^2 \omega \gamma}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (10)$$

Bessel function nodes

$J_0(x)=0$, $x = 2.405, 5.5201, 8.6537, \dots$

$J_1(x)=0$, $x = 3.833, 7.0156, 10.1735, \dots$

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a }F\left(\frac{s}{a}\right)$
Linearity	$\alpha \cdot f(x) + \beta \cdot g(x)$	$\alpha \cdot F(s) + \beta \cdot G(s)$
Shift	$f(x - a)$	$e^{-2\pi ias}F(s)$
Convolution	$f(x) * g(x)$	$F(s)G(s)$
Derivative	$\frac{d}{dx}(f(x))$	$2\pi isF(s)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx$	$\int_{-\infty}^{+\infty} F(s) ^2 ds$

Table 1: Basic theorems of Fourier theory

PHYS3114 — (more) Definitions and Formulae

Gradient	$\nabla f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$ $\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$ $\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \phi} \hat{\phi} + \frac{\partial f}{\partial z} \hat{z}$
Laplacian	$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ $\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}$ $\nabla^2 f = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$
Divergence	$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$ $\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$ $\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$
Curl	$\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z}$ $\nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \hat{r}$ $+ \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \hat{\theta} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\phi}$ $\nabla \times \mathbf{A} = \left[\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right] \hat{\phi} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right] \hat{z}$
Identities	$\nabla \times (\nabla f) = 0$ $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$
Gradient Theorem	$\int_a^b (\nabla f) \cdot d\mathbf{l} = f(\mathbf{b}) - f(\mathbf{a})$
Divergence Theorem	$\oint_S \mathbf{A} \cdot d\mathbf{a} = \int_V \nabla \cdot \mathbf{A} dv$
Stokes' Theorem	$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a}$

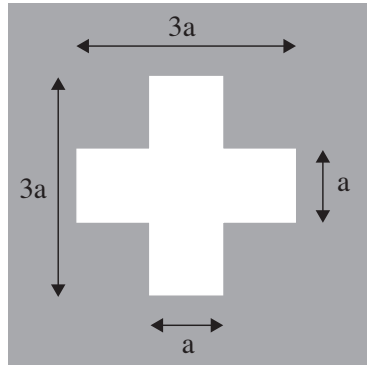
Coulomb's Law	$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$
Biot-Savart Law	$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{l} \times \hat{\mathbf{r}}}{r^2}$
Lorentz Force Law	$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$
Maxwell's Equations	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ $\nabla \cdot \mathbf{B} = 0 \quad \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$
Electric Potential	$\mathbf{E}(\mathbf{r}) = -\nabla \phi(\mathbf{r}) - \frac{\partial \mathbf{A}}{\partial t}$
Vector Potential	$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{A} = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}}{r} dv$
Charge Conservation	$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$
Current Density	$\mathbf{J} = \rho \mathbf{v} = \rho \mu \mathbf{E} = \sigma \mathbf{E}$
Poynting Vector	$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$
Stored Energy	$W = \frac{1}{2} \sum_i q_i V(\mathbf{r}_i) = \frac{1}{2} \int V(\mathbf{r}) \rho(\mathbf{r}) d^3\mathbf{r}$ $W = \frac{1}{2} \int \left(\epsilon E^2 + \frac{B^2}{\mu} \right) dv \quad W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} dv$
Faraday's Law	$\int \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi}{dt} \quad \phi = \int_S \mathbf{B} \cdot d\mathbf{a}$
Electric Displacement	$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad \int_S \mathbf{D} \cdot d\mathbf{s} = Q_f$
Magnetic Field Intensity	$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad \int_L \mathbf{H} \cdot d\mathbf{l} = I_f$
Bound Charge Densities	$\rho_b = -\nabla \cdot \mathbf{P} \quad \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$
Magnetization Current Densities	$\mathbf{J}_b = \nabla \times \mathbf{M} \quad \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$
Linear Media	$\mathbf{P} = \chi_e \epsilon_0 \mathbf{E} \quad \mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}$ $\mathbf{M} = \chi_m \mathbf{H} \quad \mathbf{B} = \mu_0 (1 + \chi_m) \mathbf{H} = \mu \mathbf{H}$
Capacitance	$C = \frac{Q}{V} \quad W = \frac{1}{2} CV^2$
Inductance	$\varepsilon = -L \frac{dI}{dt} \quad W = \frac{1}{2} LI^2$

Physical Constants

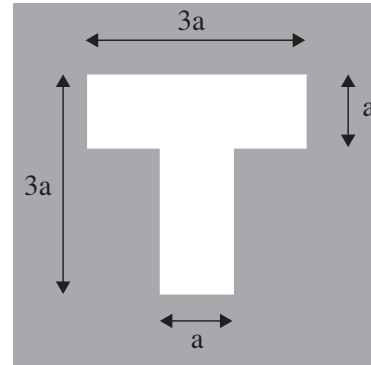
Avagadro's Number	N_A	$6.02 \times 10^{23} \text{ mole}^{-1}$
Permittivity of free space	ϵ_0	$8.85 \times 10^{-12} \text{ Fm}^{-1}$
	$\frac{1}{4\pi\epsilon_0}$	$9.0 \times 10^9 \text{ Nm}^2\text{C}^{-2}$
Permeability of free space	μ_0	$4\pi \times 10^{-7} \text{ Hm}^{-1}$
Electron charge	e	$1.6 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67 \times 10^{-27} \text{ kg}$
Speed of light in vacuum	c	$3.00 \times 10^8 \text{ ms}^{-1}$

Question 1 (20 marks)

The two apertures are depicted in the Figure below are to be used in a standard Fourier filtering application. The experimental apparatus consists of a collimated monochromatic (wavelength, λ) light source fully illuminating a thin aperture placed in front of a thin spherical lens of focal length, f . The far-field diffraction pattern is observed at the back focal plane of the lens. A second aperture is used to filter the field at the back focal plane and the transmitted field is the observed in the image plane.



APERTURE 1



APERTURE 2

For each apertures:

- Determine an expression, $f(x, y)$, in terms of spatial coordinates.
- Write an expression for the corresponding diffracted field observed in the back focal plane in terms of spatial coordinates (ζ, η) .
- A directional filter is applied to the back focal plane such that only components along the horizontal axis are transmitted. Determine the resulting field observed in the image plane for each aperture and plot the result
- The same directional filter is now rotated by 90 degrees such that only components along the vertical axis are transmitted. Determine the field in this case for each aperture, again providing a plot.

Question 2 (20 marks)

The following formula can be used to describe the intensity pattern produced by two pinholes illuminated with a monochromatic light source of finite coherence length.

$$\langle I(P) \rangle = |K_1|^2 \langle I_1 \rangle + |K_2|^2 \langle I_2 \rangle + \left[\langle I_1 \rangle^{\frac{1}{2}} \langle I_2 \rangle^{\frac{1}{2}} \right] |\gamma_{12}| \cos(\alpha_{12}(\mathbf{r}_1, \mathbf{r}_2, \tau) - \delta(\tau))$$

- (a) Explain the physical origin of each of the terms and their effect on the observed interference pattern.

You are tasked with testing the coherence properties of two different extended light sources for an interference experiment. One light source emits monochromatic light with a central wavelength of 488 nm and has a square aperture with dimensions of 1 by 1 cm. A second light source emits monochromatic light at a wavelength of 590 nm and has a circular aperture of diameter 1 cm.

- (b) Use the van Cittert - Zernike theorem to derive the complex degree of coherence for each source.

Each lamp is observed at a distance of 10 m through a narrow double slit of variable slit separation held directly against the eye.

- (c) Describe qualitatively how the observed diffraction pattern will change as a function of slit separation.
- (d) For each lamp determine the minimum slit separation for which the interference fringes will disappear.
- (e) If the eye has a focal length of approximately 22 mm, calculate the expected fringe separation corresponding to minimum fringe visibility.

Question 3 (20 marks)

Let us consider a cylindrical capacitor, which is composed of an inner cylinder (plate) with radius a enclosed by an outer cylinder (plate) with radius b . Between the plates of this cylindrical capacitor voltage V is applied. Over the inner capacitor's plate current I is passed along the cylinder's axis.

- (a) In which frame and under what conditions only magnetic field is nonzero inside the capacitor?
- (b) What is the value of this magnetic field?

Hint: You can neglect the resistance of the inner cylinder.

Question 4 (20 marks)

The Lorentz transformation from a frame S to a frame $\bar{S}$ moving with speed v along the x -axis with respect to S is described by the matrix:

$$\Lambda_{\nu}^{\mu} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (11)$$

where $\beta = v/c$, and $\gamma = 1/\sqrt{1 - \beta^2}$.

A dual to electromagnetic field strength tensor $F^{\mu\nu}$ is tensor $G^{\mu\nu}$:

$$G^{\mu\nu} = \begin{pmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & -E_z/c & E_y/c \\ -B_y & E_z/c & 0 & -E_x/c \\ -B_z & -E_y/c & E_x/c & 0 \end{pmatrix}, \quad (12)$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, and $c = 1/\sqrt{\mu_0\epsilon_0}$ is the speed of light. A second rank tensor transforms as $\bar{t}^{\mu\nu} = \Lambda_{\lambda}^{\mu}\Lambda_{\sigma}^{\nu}t^{\lambda\sigma}$, where the Einstein summation convention has been used.

- (a) Write down the individual components of the tensor $\bar{G}^{\mu\nu}$ in terms of the given components of $G^{\mu\nu}$.
- (b) Consider a point charge Q moving with speed v along the y -axis (not the x -axis) in the lab frame. Determine the electric field due to this moving charged particle in the lab frame.
- (c) What is the non-relativistic limit in part (b)?
- (d) Determine the magnetic field due to this charge and show that in the non-relativistic limit it reduces to the Biot-Savart law.

– End of Exam –